

Algorithms for sequencing data analysis



Sequencing Data Analysis

- Normalization of sequencing data
 - Same total number of reads (Ex1)
 - TMM normalization
 - Quantile normalization
- Detect rhythmical patterns using Fourier transform
- Hierarchical clustering

Advanced normalization

- Same total number of reads (Ex1)
- TMM normalization
- Quantile normalization

	Sample #1	Sample #2	Sample #3
Gene 1	200	40	700
Gene 2	300	50	650
Gene 3	150	30	900
Gene 4	180	10	850
Sum	830	130	3100
Average	207.5	32.5	775

Advanced normalization-

Same total number of reads (Ex1)

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	Sample #1	Sample #2	Sample #3
Gene 1	241	308	226
Gene 2	361	384	210
Gene 3	181	231	290
Gene 4	217	77	274
Sum	1000	1000	1000
Average	250	250	250

$$Sample1, Gene1 = \frac{200}{830} * 1000 = 241$$

Advanced normalization-

TMM normalization (Trimmed Mean of M-values)

1. Calculate the log fold change (M_i) between the two samples for each gene.
2. Compute the weights (W_i).
3. Sort the folds according to their values.
4. Trim the upper and lower percentages of the data.
5. Get the TMM after trimming and calculate the normalization factor.
6. Divide the second sample by the factor.

Genes	Sample #1	Sample #2
Gene 1	0	10
Gene 2	44	51
Gene 3	70	53
...	...	...
Gene N	16	18

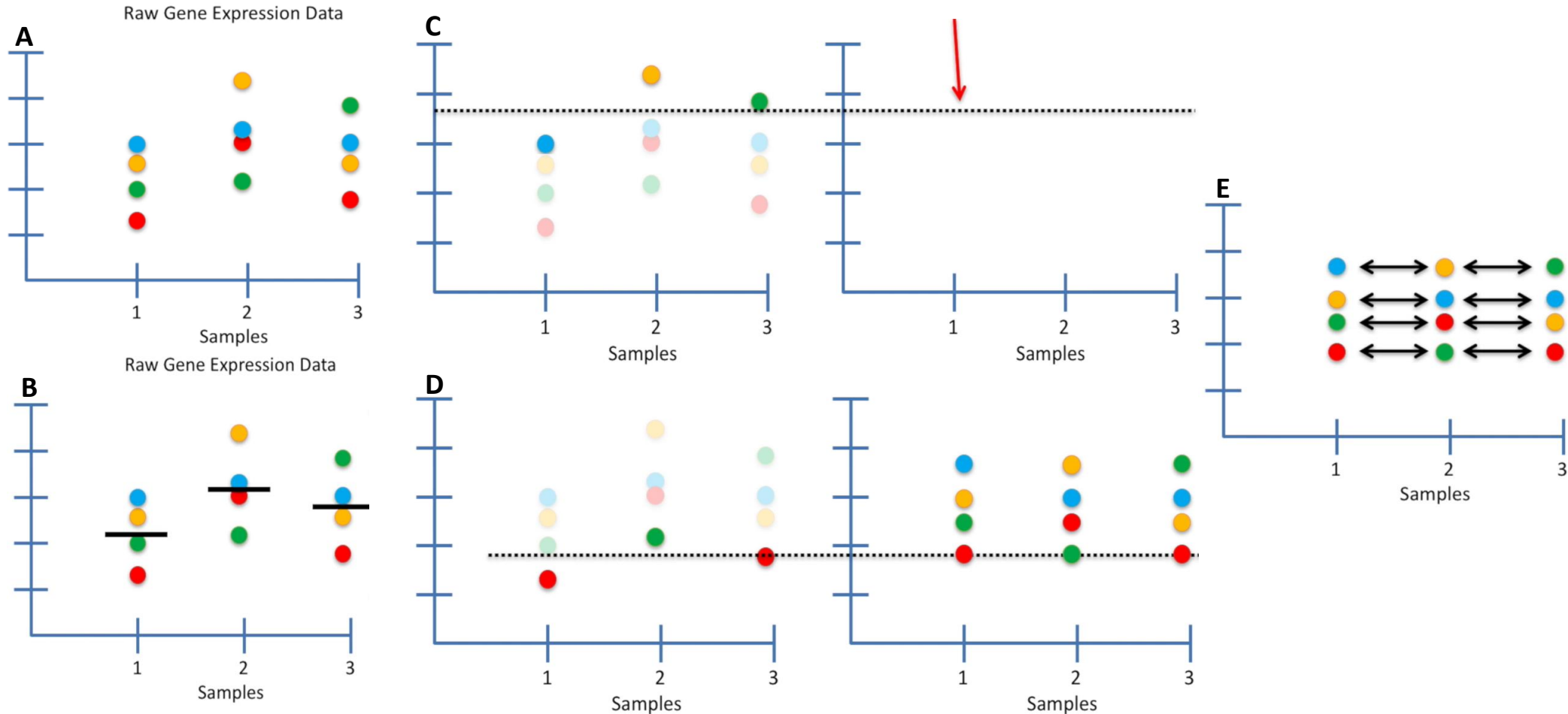
$$M_i = \log\left(\frac{\text{Sample \#2, Gene } i}{\text{Reference, Gene } i}\right)$$

$$W_i = \text{sqrt}\left(\frac{\text{Sample \#2, Gene } i + \text{Reference, Gene } i}{2}\right)$$

$$TMM = \frac{\text{sum}(W_i * M_i)}{\text{sum}(W_i)}$$

$$\text{factor} = \exp(TMM)$$

Advanced normalization- Quantile normalization



Advanced normalization-

Quantile normalization

A	Sample #1	Sample #2	Sample #3
Gene 1	200	80	50
Gene 2	300	100	30
Gene 3	50	30	300
Gene 4	150	10	180

B	Sample #1	Sample #2	Sample #3
	300	100	300
	200	80	180
	150	30	50
	50	10	30

C	Sample #1	Sample #2	Sample #3
	233	233	233
	153	153	153
	77	77	77
	30	30	30

D	Sample #1	Sample #2	Sample #3
Gene 1	153	153	77
Gene 2	233	233	30
Gene 3	30	77	233
Gene 4	77	30	153

Sort

Average

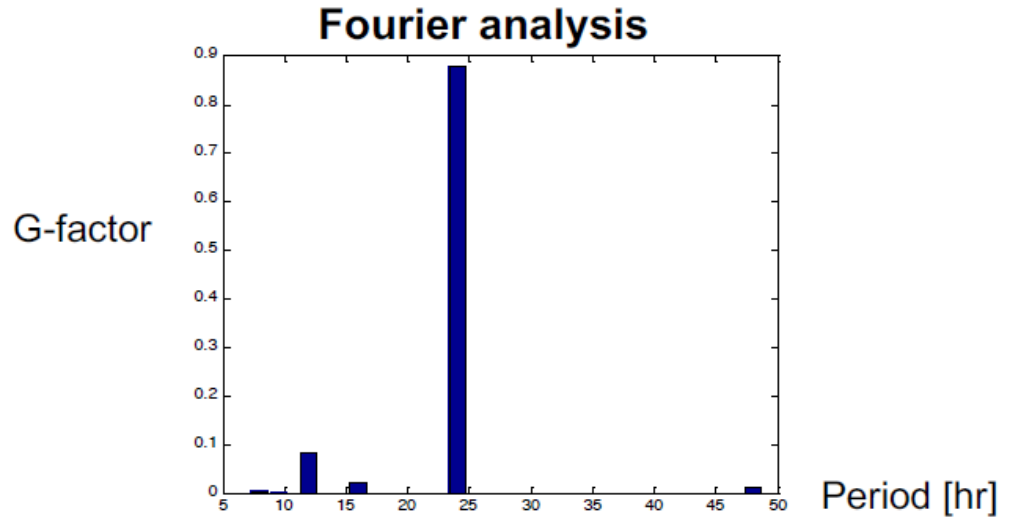
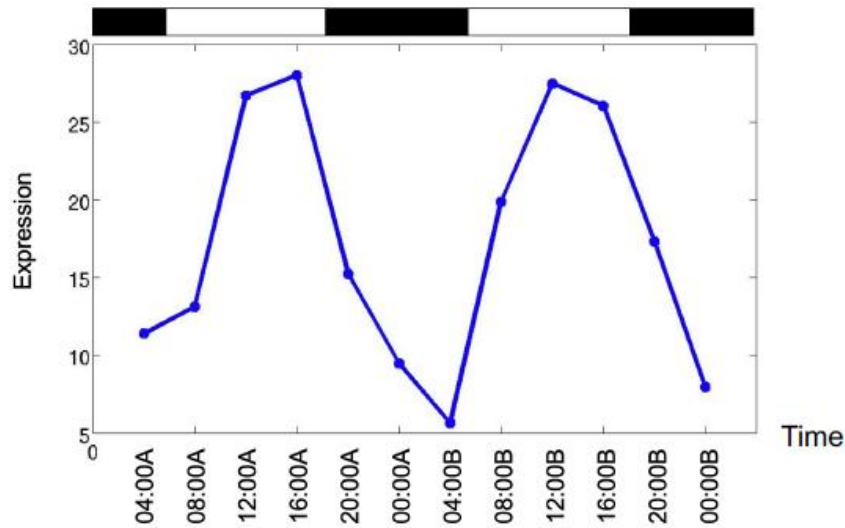
$$row1 = \frac{300 + 100 + 300}{3} = 233$$

$$row2 = \frac{200 + 80 + 180}{3} = 153$$

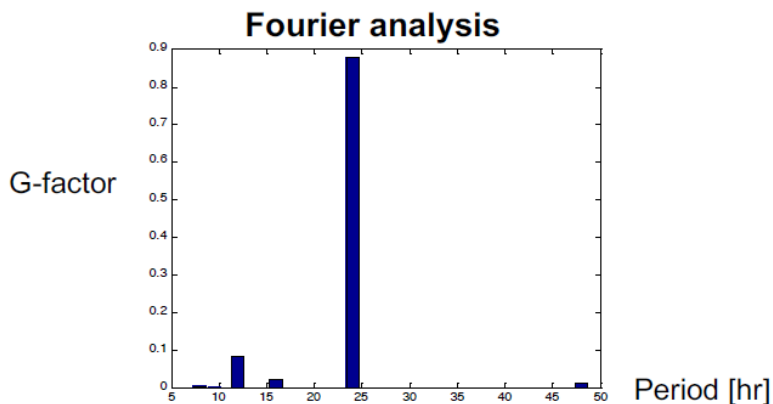
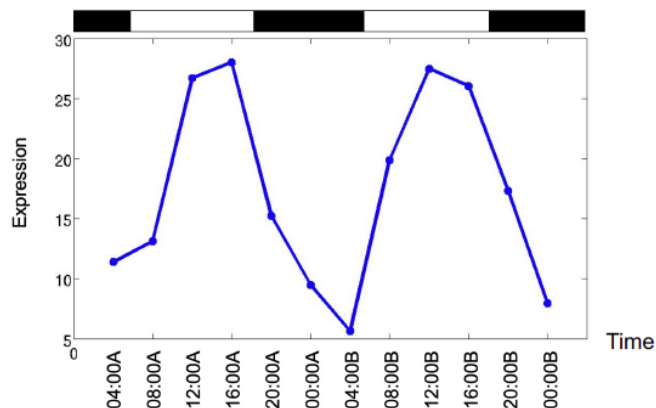
$$row3 = \frac{150 + 30 + 50}{3} = 77$$

$$row4 = \frac{50 + 10 + 30}{3} = 30$$

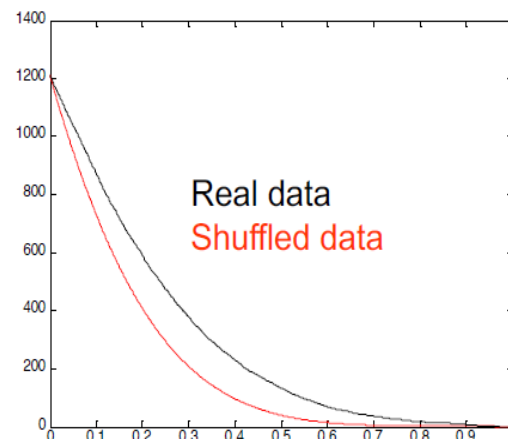
Permutations analysis to detect circadian genes with a defined true positive rate



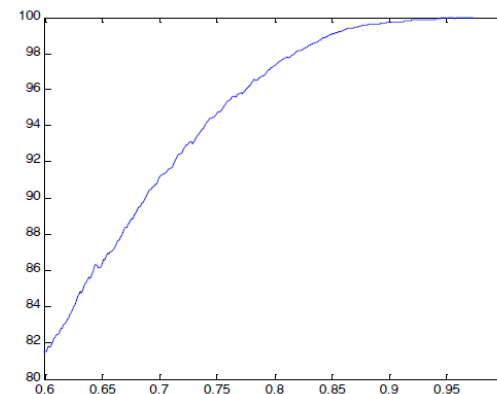
Permutations analysis to detect circadian genes with a defined true positive rate



Number of genes detected



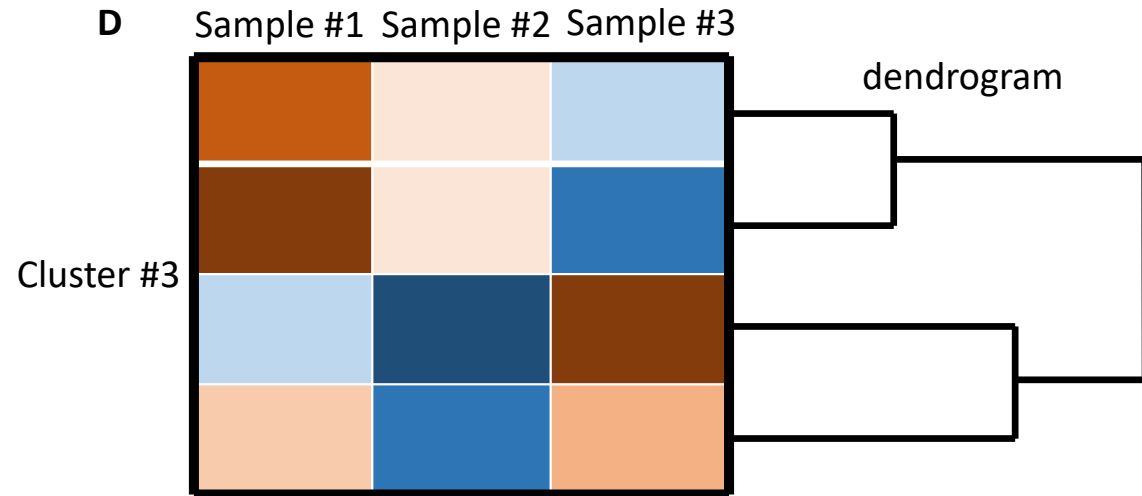
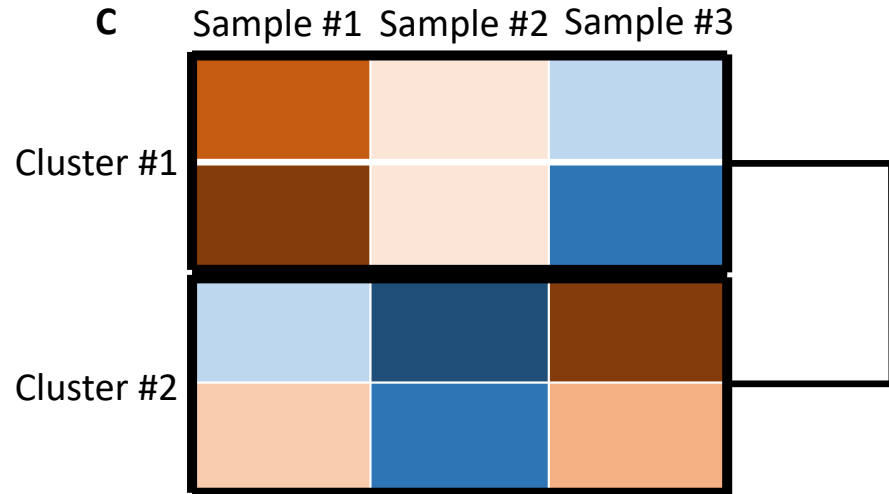
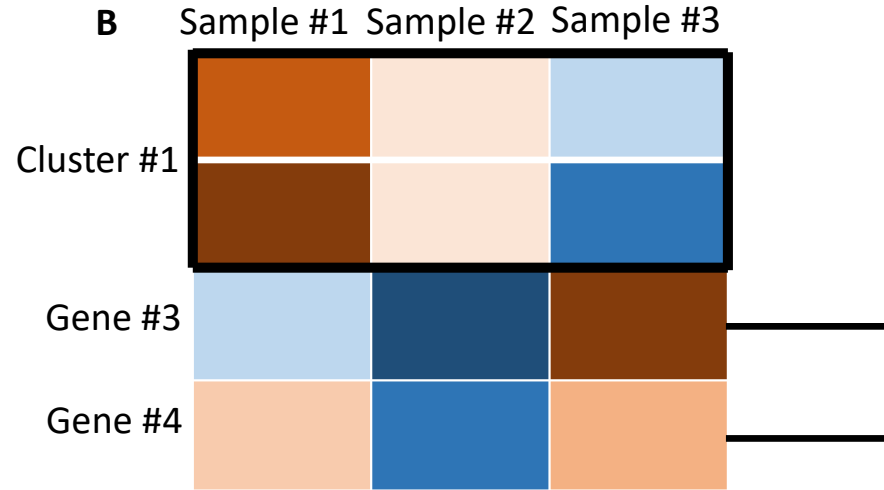
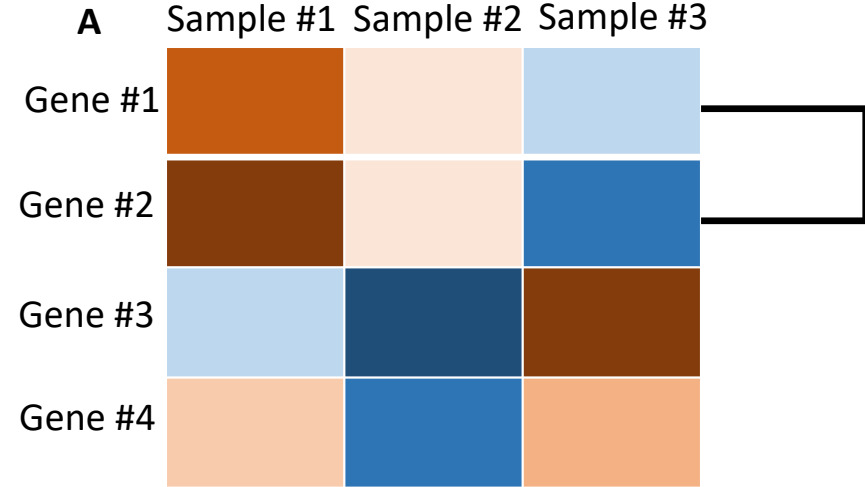
G-factor



True-Positive [%]
versus G-factor

$$TPR = \frac{TP}{P}$$

Hierarchical Clustering



Hierarchical Clustering

	Sample #1	Sample #2	Sample #3	
Gene #1	200	100	50	
Gene #2	300	100	10	
Gene #3	50	20	300	
Gene #4				

Which gene is most similar:

$$D_{12} = \sqrt{(200 - 300)^2 + (100 - 100)^2 + (50 - 10)^2} = 108$$

$$D_{13} = \sqrt{(200 - 50)^2 + (100 - 20)^2 + (50 - 300)^2} = 302$$

$$D_{23} = \dots$$

	Sample #1	Sample #2	Sample #3
Cluster #1			
Gene #3			
Gene #4			

Ways to compare to cluster:

1. Average of each cluster
2. Closest point in each cluster
3. Furthest point in each cluster

$$D_{(ij),k} = \left(\frac{n_i}{n_i + n_j}\right)D_{i,k} + \left(\frac{n_j}{n_i + n_j}\right)D_{j,k}$$